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Impact of severe infections on the risk of acute cardiovascular and cerebrovascular diseases: A prospective cohort study

Siqi Tang [a,b,c], Yonggen Jiang [d], Yiling Wu [d], Xuyan Su [d], Shuo Wang [b,e], Ruilin Chen [b,f], Genming Zhao [b,g], Wanghong Xu [b], Xing Liu [b], Ruoxin Zhang [b], Tiejun Zhang [a,b], Xingdong Chen [h,i,j], Yanfeng Jiang [k,i,*], Chen Suo [a,b,*]

[a] Shanghai Institute of Infectious Disease and Biosecurity, Fudan University, Shanghai, China
[b] Department of Epidemiology, Ministry of Education Key Laboratory of Public Health Safety, School of Public Health, Fudan University, Shanghai, China
[c] Minhang District Center for Disease Control and Prevention (Minhang District Institute of Public Health Supervision), Shanghai, China
[d] Songjiang District Center for Disease Control and Prevention (Songjiang District Institute of Public Health Supervision), Shanghai, China
[e] Yangpu District Center for Disease Control and Prevention (Yangpu District Institute of Public Health Supervision), Shanghai, China
[f] Jiangsu Hengrui Medicine Co., Ltd., Lianyungang, China
[g] Key Laboratory of Health Technology Assessment, National Health Commission of the People's Republic of China (Fudan University), Shanghai, China
[h] State Key Laboratory of Genetics and Development of Complex Phenotypes, Human Phenome Institute, Zhangjiang Fudan International Innovation Center, and National Clinical Research Center for Aging and Medicine, Huashan Hospital, Fudan University, Shanghai, China
[i] Fudan University Taizhou Institute of Health Sciences, Taizhou, Jiangsu, China
[j] Yiwu Research Institute of Fudan University, Yiwu, Zhejiang, China
[k] Human Phenome Institute, Research and Innovation Center, Shanghai Pudong Hospital, Zhangjiang Fudan International Innovation Center, Fudan University, Shanghai, China

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ABSTRACT

Background: Severe infections, particularly those requiring hospitalization, have been widely recognized as potential risk factors for cardiovascular and cerebrovascular diseases. However, the precise relationship remains unclear, particularly regarding how factors such as different time windows, repeated infections, infections caused by various pathogens, and infections involving different organ systems may influence the risk of acute cardiovascular and cerebrovascular events. This study aimed to evaluate the impact of severe infections on the incidence of these acute events by specifically focusing on these factors.

Methods: We conducted a prospective cohort study involving 55,338 participants, with a median follow-up duration of 6.42 years (IQR: 4.56–7.01). Hospitalization events related to infectious diseases and incident cardiovascular and cerebrovascular diseases were identified using passive follow-up methods. Segmented Cox regression analyses were performed to evaluate the effects of various infection-related factors on the risk of acute cardiovascular and cerebrovascular diseases, including different time windows after infection, repeated infections, infections caused by various pathogens, and infections involving different organ systems.

Results: The risk of cardiovascular and cerebrovascular events after severe infection was elevated during the whole follow-up (HR=4.07, 95% CI: 3.62–4.57) and was most significantly elevated in 3 months following severe infection (HR=8.24, 95% CI: 6.16–11.02). Repeated infections were positively correlated with the excess risk of stroke (HR=4.73, 95% CI: 1.75–12.79 for ≥3 infections, p for difference = 0.013). Infections involving different organ systems carried a much higher risk compared to single-system infections (HR= 13.11, 95% CI: 7.3–23.53). Viral infections notably increased the risk of acute ischemic heart disease (HR=4.22, 95% CI: 2.29–7.76).

Conclusion: The study found that severe infections were associated with the elevated risk of cardiovascular and cerebrovascular events. The findings suggest that more attention should be given to preventing and intervening in cardiovascular events among high-risk infection populations.

* Corresponding authors.
E-mail addresses: yanfengjiang@fudan.edu.cn (Y. Jiang), suochen@fudan.edu.cn (C. Suo).

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1. Introduction

Infection refers to inflammatory responses caused by the invasion of pathogens such as bacteria, viruses, fungi, and parasites into the human body, and infectious diseases continue to cause substantial morbidity and mortality worldwide [1,2]. For instance, lower respiratory tract infections (LRTIs) ranked as the fourth leading cause of death globally in 2019 [3]. Sepsis, a severe manifestation of infection, accounted for an estimated 49 million cases and approximately 11 million sepsis-related deaths in 2017, representing 19.7 % of all global deaths [4,5].

In addition to its acute effects, infection may also lead to longer-term health consequences. A study found that patients with a history of COVID-19 infection experienced an increased risk of infectious diseases during a one-year follow-up period [6]. The risk was particularly elevated among patients hospitalized during the acute phase, suggesting that infection, especially severe infection, may have long-term effects on the immune system. Another study found that patients hospitalized in the ICU with either septic or non-septic infections had a significantly increased risk of mortality within three years following discharge [7]. These findings collectively highlight the need to better understand the health consequences associated with severe infections. Among these consequences, increasing attention has been directed toward the association between severe infections and cardiovascular and cerebrovascular diseases [8–10].

Infection may trigger cardiovascular and cerebrovascular diseases through multiple mechanisms. For example, systemic inflammatory responses induced by infections can promote the progression of atherosclerosis and increase the risk of thrombosis [11–14]. Also, infections may also cause endothelial cell damage, leading to vascular dysfunction, thereby increasing the incidence of cardiovascular and cerebrovascular events [15]. Furthermore, cytokines and other biomarkers released during infections may activate the coagulation system, further raising the risk of thrombosis [16].

This study aims to investigate the impact of severe infections on acute cardiovascular and cerebrovascular diseases. Specifically, it seeks to analyze how cardiovascular risk evolves over different time windows following infection, examine the role of repeated infections, infection of different organ systems and pathogens, and conduct refined analyses using detailed clinical data. The goal is to provide a more accurate assessment of cardiovascular risk following severe infection, offering scientific guidance for the management and treatment of post-infection patients.

2. Methods

2.1. Study design and study population

This study utilized data from the Shanghai Suburban Adult Cohort and Biobank (SSACB), an ongoing prospective cohort study based on a natural population. The SSACB aims to investigate the characteristics and trends of chronic non-communicable diseases (NCDs) among residents and to examine the associations between environmental, lifestyle, and genetic factors and major chronic diseases. A detailed description of the SSACB methodology is available in previous publications [17]. A total of 69,116 permanent residents aged 20-74 in Songjiang, Jiading, Minhang and Xuhui districts were included in the study using multi-stage cluster random sampling. The baseline data collection was conducted from April 2016 to December 2019. The study protocol was approved by the Ethics Committee of the Fudan University, School of Public Health (IRB#2016–04–0586). Written informed consent was obtained from all participants before data collection.

The inclusion criteria for this study required participants to be from the SSACB cohort, to have provided written informed consent, and to have completed the baseline survey. Exclusion criteria included incomplete or logically inconsistent baseline questionnaires, physical examinations, or laboratory tests; a diagnosis of malignant tumors at baseline; or a history of cardiovascular or cerebrovascular diseases at baseline. The latter included participants who self-reported a history of coronary heart disease or stroke, those with records in the myocardial infarction (MI) and stroke reporting systems and those diagnosed with cardiovascular or cerebrovascular diseases by secondary or higher-level hospitals.

Among the 69,116 individuals initially enrolled, we excluded those who did not complete the baseline survey, had missing key variables, or failed the logical consistency checks of the questionnaire and physical examination ($N = 4508$). Additionally, we excluded participants with a history of malignant tumors at baseline ($N = 1270$) and those with a prior history of major cardiovascular events, including stroke, myocardial infarction, or coronary heart disease ($N = 7606$). Ultimately, 55,338 participants were included in the final analysis. (Fig. 1)

2.2. Data collection

The baseline survey was conducted by District Centers for Disease Control and Prevention (CDCs) and community health service centers. Data collection at baseline included face-to-face interviews, physical examinations, and biochemical tests. All were performed by trained investigators following standardized protocols.

The questionnaire covered various aspects, including demographic information (sex, age, educational level, marital status, etc.), lifestyle behaviors (smoking, alcohol consumption, tea consumption, physical activity), and medical history (diseases of the circulatory, endocrine, metabolic, respiratory, digestive, and urinary systems, malignant tumors, and mental disorders, etc.). Physical examination included measurements of height, weight, waist circumference, and blood pressure. Biochemical tests comprised blood lipid profiles, fasting blood glucose, and glycated hemoglobin (HbA1c), among others.

The cohort was followed up passively by integrating multiple data sources, including clinical records, the death reporting system, disease reporting systems for stroke, myocardial infarction and malignant tumor. The follow-up period for this study ended on December 31, 2023.

2.3. Exposure

Exposures were ascertained from hospital discharge diagnoses. Severe infection was defined as hospitalized infections. We used ICD-10 codes (the International Classification of Diseases, Tenth Revision) to identify infections recorded as a primary reason for a hospitalization. (Table S1) To further evaluate different infection scenarios, recurrent infection was defined as an infection occurring 14 days or more after a previous infection. Diagnoses within 14 days were considered part of the same infection episode, with the date of the first diagnosis serving as the onset time if multiple diagnoses were recorded for the same episode. In the analysis of repeated infections, the study population was categorized based on the number of infections during the follow-up period: one infection, two infections, and three or more infections. Additionally, infections were analyzed according to their primary sites, including the respiratory system, digestive system, urinary system, and multisystem infections (defined as infections involving two or more systems, such as the respiratory, digestive, urogenital, nervous, and cardiovascular systems). Pathogens were also included in the analysis, with a focus on confirmed bacterial and viral infections. We also segmented the post-infection period into distinct intervals: within 3 months after the first infection, 4–12 months, and beyond 12 months.

2.4. Outcome

We analyzed the risk of the first onset of acute ischemic heart disease (AIHD), stroke, and their combined outcome, acute cardiovascular and cerebrovascular disease (ACVD). The primary outcome data were obtained from multiple sources, including medical records, the myocardial infarction (MI) and stroke reporting systems, and the death reporting

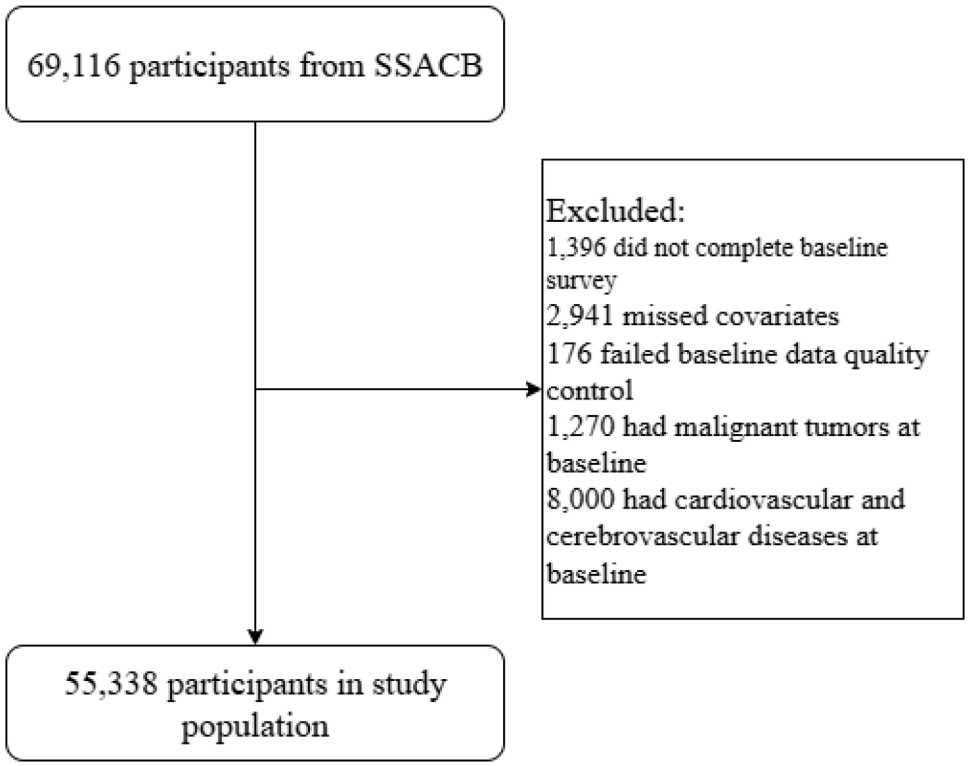
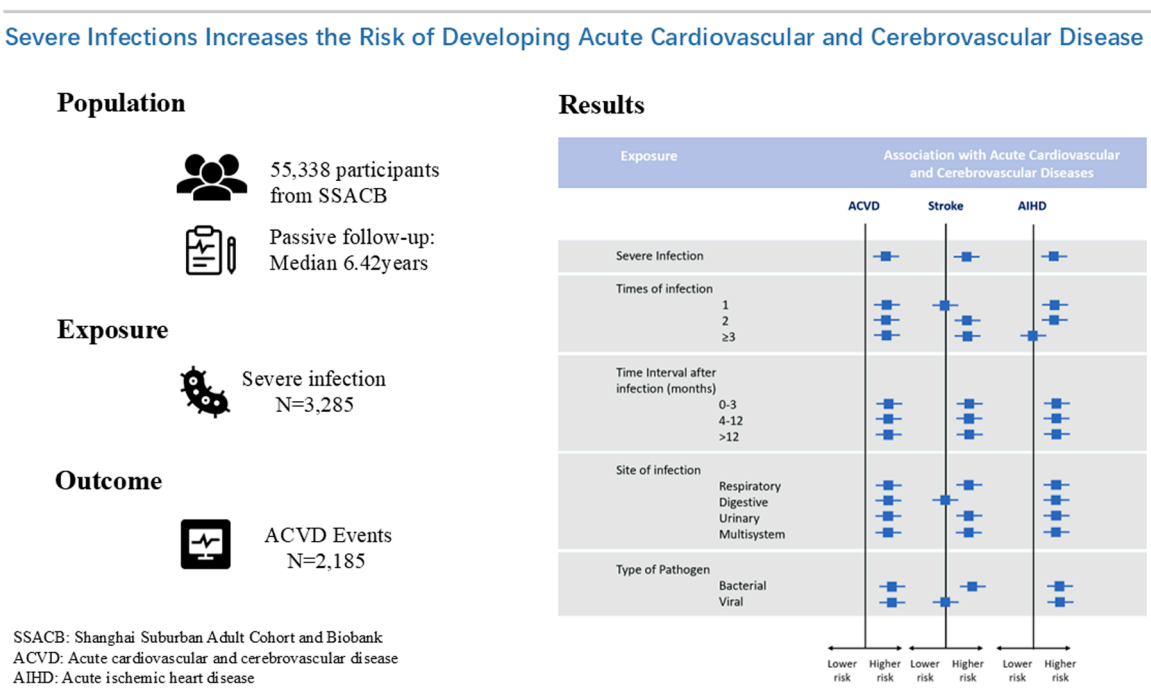


Fig. 1. Flowchart of Study Participant Selection.



Central Illustration.

system. In the medical records and death reporting system, diagnoses and causes of death were identified using ICD-10 codes. The primary outcomes of interest included stroke (I60–64) and AIHD (I20–24). For the MI and stroke reporting systems, the occurrence of stroke and AIHD was determined based on the type of report card. The time of outcome occurrence was defined as the date of the first diagnosis recorded in the electronic medical information system or the date of the first onset documented in the death reporting system, stroke reporting system, or MI reporting system.

2.5. Covariates

The covariates included basic demographic characteristics (age, sex), socioeconomic factors (education level, occupation), marital status, and

health-related demographic factors (BMI, chronic disease history, smoking, alcohol consumption, sleep quality, physical activity, etc.).

Smoking status was defined as smoking at least one cigarette per day for a continuous period of six months or longer, and participants were categorized into three groups: non-smokers, current smokers, and former smokers. Alcohol consumption was defined as drinking at least three times per week for a continuous period of six months or longer. Similarly, tea consumption was defined as drinking tea at least three times per week for a continuous period of six months or longer. BMI was categorized into four groups: normal weight (≥18.5 and <24), underweight (<18.5), overweight (≥24 and <28), and obese (≥28). Sleep quality was assessed using the Pittsburgh Sleep Quality Index (PSQI), with scores >6 indicating the presence of sleep disturbances and scores ≤5 indicating no sleep disturbances. Physical activity levels were evaluated using the International Physical Activity Questionnaire (IPAQ), and participants were classified into low, moderate, or high activity levels based on their weekly metabolic equivalent of task (MET-min/w) values. Hypertension was defined according to the *Chinese Guidelines for the Prevention and Treatment of Hypertension (2018 Revision)*, with criteria including systolic blood pressure (SBP) ≥140 mmHg and/or diastolic blood pressure (DBP) ≥90 mmHg at baseline, self-reported history of hypertension, or a relevant diagnosis recorded in a secondary or higher-level hospital. Diabetes was defined based on the *Chinese Guidelines for the Prevention and Treatment of Type 2 Diabetes (2020)*, with criteria including fasting plasma glucose (FPG) ≥7.0 mmol/L or glycated hemoglobin (HbA1c) ≥6.5 % at baseline, self-reported history of diabetes, or a relevant diagnosis recorded in a secondary or higher-level hospital. Other chronic diseases were identified through self-reported questionnaires at baseline and confirmed by relevant diagnostic records from secondary or higher-level hospitals.

2.6. Statistical analysis

In the baseline characteristics analysis, continuous variables following a normal distribution were summarized as mean ± standard deviation, and group comparisons were performed using *t*-tests or analysis of variance (ANOVA). For continuous variables with a non-normal distribution, data were presented as median (interquartile range, IQR), and the Kruskal-Wallis H test was used for comparisons. Categorical variables were expressed as frequencies (percentages), and differences between groups were assessed using the chi-square test.

To account for the time-varying nature of infection exposure, a time-dependent analysis was conducted. Participants were considered unexposed until their first infection-related hospitalization, after which they were classified as exposed. Individuals who did not experience any infection-related hospitalization during the follow-up period were considered unexposed throughout.

The association between severe infection and the incidence of acute cardiovascular and cerebrovascular events was evaluated using segmented Cox proportional hazards regression models. Exposure status was treated as a segmented variable, and hazard ratios (HRs) with 95 % confidence intervals (CIs) were calculated. Covariates, including demographic factors, were adjusted for in different models to account for potential confounding effects. To further validate the association between severe infections and the incidence of acute cardiovascular and cerebrovascular diseases, a stratified analysis was conducted based on sex, age groups, and comorbidities. Additionally, the study population was stratified using the China-PAR model [18].

All analyses were performed using R 4.2.0, and P values < 0.05 (two-sided test) were considered statistically significant.

3. Results

3.1. Cohort follow-up and demographic characteristics

Out of the 69,116 individuals in the SSACB cohort, a total of 55,338 participants were included in the final analysis. Among them, 22,727 (41.1 %) were male, and 32,611 (58.9 %) were female. The total follow-up time accumulated to 316,735.5 person-years, with a median follow-up duration of 6.42 years (IQR: 4.56–7.01 years).

The baseline characteristics of the exposure and control groups are presented in Table 1. During the follow-up period, 3285 participants experienced severe infections, while 52,053 did not. Compared to the control group, participants in the exposure group were older on average, had a higher proportion of males, and exhibited a greater prevalence of comorbidities. Additionally, they had higher rates of smoking and alcohol consumption, a greater proportion of overweight and obese individuals, lower levels of physical activity, and a higher prevalence of sleep disorders. These variables will be adjusted for in the multivariable models to account for potential confounding factors.

During the follow-up period, a total of 2485 ACVD events occurred. Among these, 350 cases were observed in the exposure group following a severe infection, while 1835 cases were recorded in the control group.

3.2. Risk of acute cardiovascular and cerebrovascular diseases after severe infection

In the fully adjusted model, severe infections were significantly associated with acute cardiovascular and cerebrovascular events (HR = 4.07, 95 % CI: 3.62–4.57; $p < 0.001$), and the HR for stroke was higher than that for AIHD (HR = 4.75, 95 % CI: 4.02–5.62 vs. HR = 3.61, 95 % CI: 3.09–4.21). Table 2

3.3. Risk analysis of repeated infections

To investigate whether severe infections have a cumulative effect, we analyzed the occurrence of repeated infections (Table 3). During the follow-up period, 3285 individuals experienced severe infections, with 2664 having one, 424 having two and 197 having three or more occurrences. Cox regression analysis revealed that, compared to individuals with no infection exposure, the adjusted HR for developing ACVD following the first severe infection was 1.73 (95 % CI: 1.42–2.10). The HR for the second infection showed no statistically significant difference from the first (2.18 [95 % CI: 1.29–3.71]; $p = 0.672$), whereas individuals with three or more infections had a significantly higher HR of 3.10 (95 % CI: 1.37–6.99) compared to the first infection ($p = 0.013$).

3.4. Risk analysis across post-infection time windows

To explore the time window effects of severe infections, we categorized the follow-up period into three intervals: 0–3 months, 4–12 months, and >12 months after the first infection (Table 4). Cox regression analysis indicated that the risk of ACVD was highest in the first 3 months following a severe infection, with an HR of 8.24 (95 % CI: 6.16–11.02). After one year, the risk decreased relative to the 0–3 months and 4–12 months intervals but remained elevated compared to the pre-infection period, with an HR of 1.65 (95 % CI: 1.33–2.03). Notably, during the 0–3 months post-infection, the HR for stroke was 16.23 (95 % CI: 10.80–24.38), which was higher than the HR for AIHD at 4.12 (95 % CI: 2.59–6.54). Table 5

3.5. Risk analysis of infections involving different organ systems

The analysis was conducted based on the classification of different organ systems, focusing on the most common severe infections: respiratory infections (1377 patients), gastrointestinal infections (653 patients), and genitourinary infections (202 patients). Patients with infections in two or more systems at the same time were defined as having multisystem infections, which included 65 individuals. The analysis revealed that infections at various sites increased the risk of ACVD, with the risk significantly heightened in patients with multisystem infections. After adjustment, the HR for multisystem infections

Table 1
Comparison of Baseline Demographic and Clinical Characteristics Between Severe Infection and Non-Severe Infection Groups.

		Total $N = 55,338$	Non-severe infection $N = 52,053$	Severe Infection $N = 3285$	P value
Age at entry, mean±SD		55.82±11.62	55.54±11.66	60.29±9.98	<0.001
Gender, n(%)	Male	22,727 (41.1)	21,195 (40.7)	1532 (46.6)	<0.001
	Female	32,611 (58.9)	30,858 (59.3)	1753 (53.4)	
Age group, n(%)	20–39	6365 (11.5)	6206 (11.9)	159 (4.8)	<0.001
	40–59	24,299 (43.9)	23,186 (44.5)	1113 (33.9)	
	60–74	24,674 (44.6)	22,661 (43.5)	2013 (61.3)	
Baseline comorbidities, n(%)					
Hypertension	No	29,268 (52.9)	27,811 (53.4)	1457 (44.4)	<0.001
	Uncontrolled	17,521 (31.7)	16,282 (31.3)	1239 (37.7)	
	Controlled	8549 (15.4)	7960 (15.3)	589 (17.9)	
Diabetes	No	48,660 (87.9)	45,978 (88.3)	2682 (81.6)	<0.001
	Uncontrolled	4892 (8.8)	4423 (8.5)	469 (14.3)	
	Controlled	1786 (3.2)	1652 (3.2)	134 (4.1)	
Respiratory disease		4925 (8.9)	4449 (8.5)	476 (14.5)	<0.001
Digestive diseases		23,279 (42.1)	21,706 (41.7)	1573 (47.9)	<0.001
Urinary diseases		8786 (15.9)	8132 (15.6)	654 (19.9)	<0.001
Mental Disorder		594 (1.1)	552 (1.1)	42 (1.3)	0.276
Other diseases		3724 (6.7)	3498 (6.7)	226 (6.9)	0.750
Smoking, n(%)	No	42,847 (77.4)	40,429 (77.7)	2418 (73.6)	<0.001
	Current smoker	10,476 (18.9)	9759 (18.7)	717 (21.8)	
	Former smoker	2015 (3.6)	1865 (3.6)	150 (4.6)	
Alcohol Assumption, n(%)		6950 (12.6)	6431 (12.4)	519 (15.8)	<0.001
Tea Assumption, n(%)		17,362 (31.4)	16,283 (31.3)	1079 (32.8)	0.064
BMI classification, n(%)	Normal weight	25,834 (46.7)	24,476 (47.0)	1358 (41.3)	<0.001
	Underweight	1603 (2.9)	1514 (2.9)	89 (2.7)	
	Overweight	21,154 (38.2)	19,832 (38.1)	1322 (40.2)	
	Obesity	6747 (12.2)	6231 (12.0)	516 (15.7)	
Central obesity, n(%)		11,342 (20.5)	10,488 (20.1)	854 (26.0)	<0.001
Physical activity, n(%)	Light	32,884 (59.4)	30,851 (59.3)	2033 (61.9)	0.004
	Moderate	15,421 (27.9)	14,536 (27.9)	885 (26.9)	
	Vigorous	7033 (12.7)	6666 (12.8)	367 (11.2)	
Sleeping Disorder, n(%)		15,191 (27.5)	14,225 (27.3)	966 (29.4)	0.010

Table 2
Results of Time-Dependent Cox Regression Analysis for Severe Infection and Risk of Acute Cardiovascular and Cerebrovascular Diseases.

Outcome	Model0		Model1		Model2	
	HR (95 % CI)	P value	HR (95 % CI)	P value	HR (95 % CI)	P value
ACVD	5.26(4.70–5.89)	<0.001	4.23(3.78–4.75)	<0.001	4.07(3.62–4.57)	<0.001
Stroke	6.32(5.37–7.44)	<0.001	4.96(4.21–5.85)	<0.001	4.75(4.02–5.62)	<0.001
AIHD	4.58(3.94–5.33)	<0.001	3.75(3.22–4.37)	<0.001	3.61(3.09–4.21)	<0.001

Segmented Cox Regression models were used. Model 0 was unadjusted. Model 1 adjusted for age and sex. Model 2 adjusted for comorbidities, lifestyle factors, and body mass index (BMI).

was 13.11 (95 % CI: 7.3-23.53). The effect on stroke (HR: 25.65, 95 % CI: 13.38-49.18) was more pronounced than that on acute ischemic heart disease (HR: 25.65, 95 % CI: 13.38-49.18).

3.6. Risk analysis of infections caused by various pathogens

Since infections were categorized based on ICD-10 codes, a total of 1047 patients were identified with bacterial infections, while 306 with viral infections, with no patients showing mixed infections. Bacterial infections were found to increase the risk of both stroke and AIHD, with a combined HR of 2.64 (95 % CI: 2.00-3.47). Viral infections posed a higher risk for AIHD, with a HR of 4.22 (95 % CI: 2.29-7.76), while the risk for stroke showed no statistically significant difference ($p = 0.958$). However, fewer than 50 % of the medical records specified the infectious pathogen type through ICD-10 codes, which may limit the robustness of the findings. (Table S2)

3.7. Subgroup and interaction analyses

Subgroup and interaction analyses were performed based on sex, age, comorbidity status, and baseline disease risk. In the sex-stratified analysis, we observed that after adjusting for covariates, the HR for stroke was higher in females than in males (6.41 [95 % CI: 5.10-8.06] vs. 3.70 [95 % CI: 2.91-4.71]), while the HR for AIHD was lower in females compared to males (2.96 [95 % CI: 2.34-3.74] vs. 4.31 [95 % CI: 3.51-5.31]) (Table S3). In the age-stratified analysis, no significant interaction effects were observed for any of the outcomes, with ACVD ($P = 0.342$), stroke ($P = 0.144$), and AIHD ($P = 0.818$) (Table S4). When stratifying by comorbidity status, we found that patients with ≤1 comorbidity had a higher HR for AIHD compared to those with ≥2 comorbidities (5.55 [4.33-7.11] vs. 3.53 [2.91-4.28], P for interaction = 0.032) (Table S5). Interaction analysis based on the China-PAR CVD risk stratification tool revealed that all interaction P-values were greater than 0.05, indicating no statistically significant differences between the groups (Table S6).

3.8. Sensitivity analysis

To assess the robustness of the association between acute-phase infection and the incidence of ACVD, we performed a sensitivity analysis excluding patients who developed ACVD within 7 days of the severe infection. This analysis showed a slight attenuation of the effect (Table S7). However, in a time-segment analysis, we found that the effect for ACVD within the first 3 months significantly decreased

Table 3
Cox Regression Analysis for the Impact of Severe Infection Frequency on Acute Cardiovascular and Cerebrovascular Diseases.

Outcome	Exposure	Model 0			Model 1			Model 2		
		HR (95 % CI)	P value	P for difference	HR (95 % CI)	P value	P for difference	HR (95 % CI)	P value	P for difference
ACVD	First infection	2.15 (1.77–2.61)	<0.001	Ref	1.77 (1.46–2.15)	<0.001	Ref	1.73 (1.42–2.10)	<0.001	Ref
	Second infection	2.82 (1.66–4.79)	<0.001	0.605	2.26 (1.33–3.82)	0.002	0.672	2.18 (1.29–3.71)	0.004	0.672
	Third and more infections	3.84 (1.71–8.60)	0.001	0.006	3.16 (1.40–7.14)	0.006	0.029	3.10 (1.37–6.99)	0.006	0.029
Stroke	First infection	1.76 (1.26–2.46)	<0.001	Ref	1.40 (1.00–1.96)	0.050	Ref	1.38 (0.98–1.93)	0.062	Ref
	Second infection	2.91 (1.30–6.51)	0.009	0.496	2.28 (1.02–5.08)	0.044	0.508	2.24 (1.01–4.98)	0.047	0.508
	Third and more infections	6.02 (2.25–16.10)	<0.001	0.002	4.74 (1.74–12.94)	0.002	0.012	4.73 (1.75–12.79)	0.002	0.013
AIHD	First infection	2.41 (1.90–3.05)	<0.001	Ref	2.03 (1.60–2.58)	<0.001	Ref	1.97 (1.55–2.49)	<0.001	Ref
	Second infection	2.76 (1.38–5.54)	0.004	0.929	2.23 (1.11–4.48)	0.024	0.966	2.13 (1.06–4.29)	0.034	0.966
	Third and more infections	2.23 (0.55–8.97)	0.261	0.674	1.88 (0.46–7.59)	0.377	0.814	1.82 (0.45–7.40)	0.404	0.814

Segmented Cox Regression models were used. Model 0 was unadjusted. Model 1 adjusted for age and sex. Model 2 adjusted for comorbidities, lifestyle factors, and body mass index (BMI).

Table 4
Cox Regression Analysis for the Impact of Severe Infection at Different Time Intervals on Acute Cardiovascular and Cerebrovascular Diseases.

Outcome	Time after infection	Model 0		Model 1		Model 2	
		HR (95 % CI)	P value	HR (95 % CI)	P value	HR (95 % CI)	P value
ACVD	0–3 months	11.00(8.23–14.71)	<0.001	8.61(6.44–11.52)	<0.001	8.24(6.16–11.02)	<0.001
	4–12 months	2.87(2.01–4.11)	<0.001	2.29(1.60–3.28)	<0.001	2.19(1.53–3.14)	<0.001
	>12 months	2.07(1.68–2.56)	<0.001	1.72(1.39–2.12)	<0.001	1.65(1.33–2.03)	<0.001
Stroke	0–3 months	22.47(14.97–33.73)	<0.001	17.08(11.37–25.66)	<0.001	16.23(10.80–24.38)	<0.001
	4–12 months	2.93(1.68–5.11)	<0.001	2.27(1.30–3.97)	0.004	2.16(1.24–3.77)	0.007
	>12 months	2.07(1.48–2.89)	<0.001	1.67(1.19–2.34)	0.003	1.60(1.14–2.24)	0.006
AIHD	0–3 months	5.38(3.39–8.53)	<0.001	4.29(2.70–6.81)	<0.001	4.12(2.59–6.54)	<0.001
	4–12 months	2.83(1.77–4.52)	<0.001	2.30(1.44–3.67)	<0.001	2.21(1.38–3.52)	0.001
	>12 months	2.08(1.59–2.71)	<0.001	1.75(1.34–2.29)	<0.001	1.67(1.28–2.19)	<0.001

Segmented Cox Regression models were used. Model 0 was unadjusted. Model 1 adjusted for age and sex. Model 2 adjusted for comorbidities, lifestyle factors, and body mass index (BMI).

Table 5
Cox Regression Analysis for Site of Infection of Severe Infection on Acute Cardiovascular and Cerebrovascular Diseases.

Site of infection	Outcome	Model 0		Model 1		Model 2	
		HR (95 % CI)	P value	HR (95 % CI)	P value	HR (95 % CI)	P value
Respiratory	ACVD	2.92(2.32–3.69)	<0.001	2.25(1.78–2.84)	<0.001	2.23(1.77–2.82)	<0.001
	Stroke	3.37(2.40–4.73)	<0.001	2.51(1.79–3.53)	<0.001	2.55(1.81–3.60)	<0.001
	AIHD	2.63(1.92–3.60)	<0.001	2.06(1.50–2.82)	<0.001	2.00(1.46–2.74)	<0.001
Digestive	ACVD	2.56(1.37–4.79)	0.003	2.28(1.23–4.23)	0.009	2.30(1.24–4.26)	0.008
	Stroke	1.29(0.32–5.18)	0.718	1.10(0.28–4.37)	0.891	1.14(0.29–4.55)	0.851
	AIHD	3.40(1.68–6.87)	<0.001	3.11(1.55–6.26)	0.001	3.07(1.52–6.19)	0.002
Urinary	ACVD	2.20(1.49–3.25)	<0.001	1.88(1.27–2.80)	0.002	1.77(1.19–2.64)	0.005
	Stroke	2.30(1.27–4.18)	0.006	1.95(1.07–3.55)	0.030	1.87(1.02–3.41)	0.043
	AIHD	2.13(1.28–3.55)	0.004	1.84(1.10–3.08)	0.020	1.71(1.02–2.87)	0.041
Multisystem	ACVD	18.88(10.62–33.57)	<0.001	14.29(7.91–25.82)	<0.001	13.11(7.3–23.53)	<0.001
	Stroke	35.91(18.73–68.86)	<0.001	28.16(14.51–54.64)	<0.001	25.65(13.38–49.18)	<0.001
	AIHD	8.19(2.96–22.69)	<0.001	6.06(2.15–17.06)	<0.001	5.60(1.96–15.97)	0.001

Segmented Cox Regression models were used. Model 0 was unadjusted. Model 1 adjusted for age and sex. Model 2 adjusted for comorbidities, lifestyle factors, and body mass index (BMI).

compared to the analysis including all patients, suggesting that the increased risk of ACVD within 3 months post-infection is primarily driven by the acute effect occurring within the first 7 days (Table S8).

Additionally, we conducted a competing risks analysis using the Fine-Gray model (Table S9). Considering the competing risk of mortality, the results suggest that severe infections still significantly promote the occurrence of ACVD.

4. Discussion

4.1. Prior research and summary of key findings

A lot of evidence has demonstrated a close association between infections and cardiovascular and cerebrovascular events [8–10,19–21]. Over time, research in this field has evolved from ecological

observations to more rigorous epidemiological designs. In 1978, an ecological study by Bainton et al. reported increased ischemic heart disease mortality across all age groups during influenza outbreaks, suggesting acute infection as a potential trigger for myocardial infarction [22]. This study offered an early indication of the link between infection and cardiovascular events. A 1988 case-control study identified significantly elevated antibody titers against a specific *Chlamydia* species in patients with acute myocardial infarction and chronic coronary heart disease, pointing toward a possible role of bacterial infection in atherogenesis [23]. A 2004 self-controlled case series analysis demonstrated a 4.95-fold increase in myocardial infarction risk and a 3.19-fold increase in stroke risk within three days of respiratory infection diagnosis [8]. More recently, a 2020 cohort study found that individuals with community-acquired bacteremia had a 25.5-fold increased risk of myocardial infarction or stroke within 30 days post-infection, compared to the general population, which highlights the vascular risk associated with severe infections [24].

Our study found out that severe infections were associated with approximately a fourfold increased risk of acute cardiovascular and cerebrovascular events compared to individuals without such infections. The risk was most significantly elevated within 3 months after infection, gradually declining over time but remaining consistently elevated relative to those without severe infections. Individuals who experienced more than three infections exhibited a markedly increased risk, suggesting a potential dose–response relationship. Infections affecting different organ systems were all associated with an increased risk of acute cardiovascular and cerebrovascular events, with the highest risk observed in cases involving multiple systems. The impact of different pathogens on the risk varied, and differences were also observed across population subgroups. Men showed a higher risk of acute ischemic heart disease following severe infection, while women were more likely to experience stroke. Individuals with fewer pre-existing comorbidities demonstrated a relatively higher risk increase. No significant interaction was found between severe infection and either age or the China-PAR score in relation to cardiovascular and cerebrovascular risk.

4.2. Potential explanations for observed associations

Repeated infections may cause cumulative damage to the immune and vascular systems, leading to a chronic inflammatory state and endothelial dysfunction [25]. Additionally, repeated infections may reflect poorer baseline health or a decline in immune function, as well as reduced vascular repair and anti-inflammatory capacities, which further increases the risk of ACVD.

During the acute phase post-infection, cytokine storms, endothelial injury, and the activation of systemic inflammatory responses may directly lead to vasoconstriction, hemodynamic changes, and thrombosis, all of which place significant strain on the cardiovascular system in a short period [26,27]. Moreover, the chronic low-grade inflammation induced by infections may persist, keeping the cardiovascular system in a sustained subclinical injury state. Although the high risk during the acute phase may diminish over time, long-term cardiovascular monitoring and intervention following an infection may remain [28,29].

Significantly elevated risk brought by multisystem infections is possibly due to the fact that multisystem infections typically involve more severe systemic inflammation and immune system activation, leading to widespread endothelial damage and a hypercoagulable state [30]. Compared to single-system infections, multisystem infections may release higher levels of inflammatory mediators and are often accompanied by more severe physiological disturbances, which exacerbate coagulation abnormalities [12].

Different pathogens exhibited distinct characteristics. Bacterial infections increased the risks of both stroke and AIHD, whereas viral infections notably raised only the risk of AIHD to a greater extent than bacterial infections [31,32]. Bacterial infections trigger acute systemic inflammation, which promotes thrombosis and endothelial damage, thus increasing the risk of both cerebrovascular and cardiovascular diseases. In contrast, viral infections might contribute to AIHD by causing myocarditis and increasing cardiac workload, thereby exacerbating coronary blood flow insufficiency and promoting the development of AIHD [33–35]. However, we acknowledge that the validity of our findings may be limited, as fewer than 50 % of the medical records used ICD-10 coding to specify the pathogen type.

In stratified analysis, we observed that the risk of stroke was higher in women than in men, while the risk of AIHD was lower in women. This gender difference may be related to gender-specific physiological and immune response mechanisms [36–38]. Interestingly, patients with fewer comorbidities had a greater risk of developing AIHD after a severe infection compared to those with more comorbidities. This may be because patients with fewer comorbidities exhibit a more intense immune response, which leads to insufficient cardiovascular adaptation. Alternatively, it could be due to the more stringent cardiovascular monitoring and management of patients with multiple comorbidities, which reduces the occurrence of acute heart disease. Although age and CVD risk stratification are typically important factors influencing ACVD risk [39], we did not observe any interaction between age stratification, CVD risk, and the occurrence of ACVD after severe infection in this study. This might be because the trends in the occurrence of cardiovascular events after infection were consistent across different age groups and CVD risk levels.

4.3. Strengths and limitations

One of the strengths of this study lies in its use of a cohort population, combined with data from multiple sources, including baseline cohort data, medical records, stroke and myocardial infarction registry data, and mortality registry data. This comprehensive approach allowed for a thorough assessment of the risk of severe infections and ACVD, taking into account factors such as repeated infections, infection intervals, infection sites, and the pathogens involved.

This study has several limitations. First, as the study population was entirely derived from the SSACB and was without a history of without a history of cardiovascular diseases, the generalizability of the findings may be restricted due to limited population characteristics, sample sizes, and follow-up durations. Second, not all relevant confounding factors, such as genetic profiles, inflammatory levels, or infection treatment pattern and adherence, were available, which may affect the accuracy of risk estimation. Third, the scope of the study was restricted to severe infections and major ACVD events, without considering mild infections or a broader range of cardiovascular conditions, thereby limiting the comprehensiveness and mechanistic insights of the findings. Additionally, patients hospitalized for severe infections are likely to receive more intensive monitoring, which could increase the probability of detecting acute cardiovascular or cerebrovascular events. Nonetheless, as our study was restricted to clinically overt outcomes, the risk of substantial outcome misclassification is expected to be limited.

4.4. Future directions

Future research should emphasize systematic collection of pathogen-specific information, incorporate genetic and inflammatory factors at baseline into risk models, and include repeated measurements of immune biomarkers after severe infection, as well as analyses of intermediate cardiovascular biomarkers in relation to relapse frequency, to better clarify underlying biological mechanisms.

It is also important to note that our study focused exclusively on the impact of severe infection on the risk of ACVD events in individuals without a history of CVD. Future studies should extend this line of inquiry to populations with pre-existing chronic CVD, such as coronary heart disease, or a history of major acute events, such as myocardial infarction or stroke, to examine whether severe infection similarly influences the risk of recurrent cardiovascular events in these more

vulnerable groups.

Furthermore, the results should be evaluated in broader and more diverse populations to confirm their validity across different demographic and clinical settings. Future research should validate these findings by computing the risk score after incorporating all relevant components in diverse populations.

5. Conclusion

Our findings suggest that a history of severe infection should be considered a risk factor for acute cardiovascular and cerebrovascular events. Early screening and preventive interventions for cardiovascular events should be strengthened in patients with severe infections. Future research could further explore the cardiovascular risk differences among broader population with different types of infections and underlying health conditions, while also incorporating inflammatory markers. In high-risk infection groups, more attention may need to be given to the prevention and intervention of cardiovascular events.

Data availability

The datasets generated and analyzed during the current study are not publicly available due to the data of the SSACB cohort not being open access, but are available from the corresponding author on reasonable request.

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Ethics approval and consent to participate

SSACB was constructed under the approval of the ethical review board of the School of Public Health of Fudan University (IRB#2016-04-0586). Written informed consent was obtained from all participants of the SSACB cohort.

CRediT authorship contribution statement

Siqi Tang: Writing – original draft, Formal analysis. **Yonggen Jiang:** Project administration. **Yiling Wu:** Project administration. **Xuyan Su:** Project administration. **Shuo Wang:** Data curation. **Ruilin Chen:** Data curation. **Genming Zhao:** Writing – review & editing. **Wanghong Xu:** Writing – review & editing. **Xing Liu:** Writing – review & editing. **Ruoxin Zhang:** Writing – review & editing. **Tiejun Zhang:** Writing – review & editing. **Xingdong Chen:** Writing – review & editing. **Yanfeng Jiang:** Writing – review & editing, Supervision. **Chen Suo:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ajpc.2026.101436.

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